

Markscheme

Mathematics: analysis and approaches
Practice question bank

55. (a)

Consider the simultaneous equations $2^x \times 4^y = 32$ and $8^x \times 2^y = 64$.

Find the value of x and the value of y .

express both equations with base 2: $x + 2y = 5$ and $3x + y = 6$

(M1) A1 A1

substitute $x = 5 - 2y$ into the second equation: $3(5 - 2y) + y = 6$

(M1)

$$15 - 5y = 6 \Rightarrow y = 1.8$$

A1

$$x = 5 - 2(1.8) = 1.4$$

A1

[6 marks]